Mathematics Notes for Scientific Computing

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Chapter 1

Algebra in Prefix Notation

Write operations in prefix form: $(\cdot\ a\ b)$ instead of $a \cdot b$ and $(+\ a\ b)$ instead of $a + b$. A group is $(G, \cdot, e, ^{-1})$ with axioms

$$(=\ (\cdot\ e\ a)\ a), \quad (=\ (\cdot\ a\ e)\ a), \quad (=\ (\cdot\ a\ (a^{-1}))\ e), \quad (=\ (\cdot\ (\cdot\ a\ b)\ c)\ (\cdot\ a\ (\cdot\ b\ c))).$$

A ring has an abelian group $(R, +, 0, -)$, a monoid $(R, \cdot, 1)$, and distributivity:

$$(=\ (\cdot\ a\ (+\ b\ c))\ (+\ (\cdot\ a\ b)\ (\cdot\ a\ c))).$$

A field is a commutative ring where every nonzero element has a multiplicative inverse. A vector space over F has $+ : V^2 \to V$ and scalar multiplication $F \times V \to V$ satisfying linearity. An algebra adds a bilinear product $V \times V \to V$. A lattice has $\wedge, \vee$ with commutativity, associativity, absorption, and idempotence.

Chapter 2

Analysis and PDE Notation

For a field $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}^m$, use coordinate derivatives $\partial_i u_\alpha = \frac{\partial u_\alpha}{\partial x_i}$. A conservation law is

$$\partial_t u_\alpha + \partial_i f_{\alpha i}(u, x, t) = s_\alpha.$$

Poisson's equation and weak form are

$$-\partial_i(k\partial_i u) = f, \qquad \int_\Omega k\,\partial_i u\,\partial_i v\,dx = \int_\Omega f v\,dx.$$

The heat equation is $\partial_t u - \partial_i(\kappa\partial_i u) = f$. Linear elasticity uses strain $\epsilon_{ij}(u) = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ and stress $\sigma_{ij} = C_{ijkl}\epsilon_{kl}$ with equilibrium $-\partial_i\sigma_{ij} = b_j$.

Chapter 3

Geometry

A smooth manifold M is locally modeled by charts $\phi : U \subset M \to \mathbb{R}^n$. A vector field is a derivation on smooth functions. A connection ∇ differentiates vector fields with

$$\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k.$$

Curvature measures non-commuting covariant derivatives:

$$R(\partial_i, \partial_j)\partial_k = R^\ell{}_{kij}\partial_\ell, \quad R^\ell{}_{kij} = \partial_i\Gamma^\ell_{jk} - \partial_j\Gamma^\ell_{ik} + \Gamma^\ell_{im}\Gamma^m_{jk} - \Gamma^\ell_{jm}\Gamma^m_{ik}.$$

For a metric g_{ij}, the Levi-Civita connection is torsion-free and metric compatible:

$$\Gamma^k_{ij} = \frac{1}{2}g^{k\ell}(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

Chapter 4

Probability

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$ with countably additive measure $\mathbb{P}(\Omega) = 1$. A random variable is a measurable map $X : \Omega \to S$. Expectation is $\mathbb{E}[X] = \int_\Omega X(\omega) d\mathbb{P}(\omega)$. A stochastic process is a family X_t. An Itô diffusion satisfies

$$dX_t^i = b_i(X_t, t)dt + \sigma_{ij}(X_t, t)dW_t^j.$$

Itô's lemma for $f(X_t, t)$ is

$$df = \left(\partial_t f + b_i \partial_i f + \frac{1}{2} a_{ij} \partial_i \partial_j f\right) dt + \partial_i f\, \sigma_{ij} dW_t^j, \quad a = \sigma\sigma^T.$$

The Fokker–Planck PDE for density $p(x, t)$ is

$$\partial_t p = -\partial_i(b_i p) + \frac{1}{2} \partial_i \partial_j (a_{ij} p).$$